THREE EQUIVALENT CONDITIONS FOR BASE CHANGE OF A SINGLE COMPLEX IN A CARTESIAN SQUARE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For U, U' as in condition (2) there is a canonical equality of derived tensor products RGamma(U, K) tensor O_{X'}(U') = RGamma(U', L(g')^*K), so the second arrow of (2) is induced by the given map L(g')^*K to K'. used at proof lines 6686-6691. Identifies the right hand arrow of the composition in (2) with pullback followed by the given map.
- The first arrow of the composition in (2) is the base change comparison map of More on Algebra for the affine opens involved. invoked at proof lines 6691-6693. Identifies the left hand arrow with the standard comparison map whose properties are established in More on Algebra.
- A map of modules over a ring is an isomorphism if and only if all localizations at primes are isomorphisms, in the form used to compare a module map with its stalkwise versions. invoked at proof lines 6740-6746. Converts the equality of the displayed localized expressions into the equivalence of conditions (1) and (2).

Source reference. Original equation 0DJ7: sources/stacks-perfect/pinned-original-source.tex, source lines 6644–6644. Referenced at statement line 6664 in item (1); the displayed equation lies just before the statement and is not part of the counted span.

Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Cited at proof lines 6686-6691 for the equality RGamma(U, K) tensor O_{X'}(U') = RGamma(U', L(g')^*K); the identification of the arrow is inside the counted span.

Source reference. Original equation 065Z: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof line 6693 as the identity of the first arrow of (2); the identification step is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HN: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof line 6746 to conclude the equivalence of the cohomology condition with the stalkwise condition (1); the localization computation it is applied to is inside the counted span.

Lemma 0.2. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram of schemes. Let $K \in D_{QCoh}(\mathcal{O}_X)$ and let $L(g')^*K \to K'$ be a map in $D_{QCoh}(\mathcal{O}_{X'})$. The following are equivalent

1. *for any $x' \in X'$ and $i \in \mathbf{Z}$ the map (0DJ7) is an isomorphism,*
2. *for $U \subset X$, $V' \subset S'$ affine open both mapping into the affine open $V \subset S$ with $U' = V' \times_V U$ the composition*
$$R\Gamma(U, K) \otimes^{\mathbf{L}}_{\mathcal{O}_S(U)} \mathcal{O}_{S'}(V') \to R\Gamma(U, K) \otimes^{\mathbf{L}}_{\mathcal{O}_X(U)} \mathcal{O}_{X'}(U') \to R\Gamma(U', K')$$
is an isomorphism in $D(\mathcal{O}_{S'}(V'))$, and
3. *there is a set I of quadruples U_i, V_i', V_i, U_i', $i \in I$ as in (2) with $X' = \bigcup U_i'$.*

Proof. The second arrow in (2) comes from the equality

$$R\Gamma(U, K) \otimes^{\mathbf{L}}_{\mathcal{O}_X(U)} \mathcal{O}_{X'}(U') = R\Gamma(U', L(g')^*K)$$

of Lemma 08DW and the given arrow $L(g')^*K \to K'$. The first arrow of (2) is More on Algebra, Equation (065Z). It is clear that (2) implies (3). Observe that (1) is local on X'. Therefore it suffices to show that if X, S, S', X' are affine, then (1) is equivalent to the condition that

$$R\Gamma(X, K) \otimes^{\mathbf{L}}_{\mathcal{O}_S(S)} \mathcal{O}_{S'}(S') \to R\Gamma(X, K) \otimes^{\mathbf{L}}_{\mathcal{O}_X(X)} \mathcal{O}_{X'}(X') \to R\Gamma(X', K')$$

is an isomorphism in $D(\mathcal{O}_{S'}(S'))$. Say $S = \mathrm{Spec}(R)$, $X = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(R')$, $X' = \mathrm{Spec}(A')$, K corresponds to the complex $M^\bullet$ of A-modules, and K' corresponds to the complex $N^\bullet$ of A'-modules. Note that $A' = A \otimes_R R'$. The condition above is that the composition

$$M^\bullet \otimes^{\mathbf{L}}_R R' \to M^\bullet \otimes^{\mathbf{L}}_A A' \to N^\bullet$$

is an isomorphism in $D(R')$. Equivalently, it is that for all $i \in \mathbf{Z}$ the map

$$H^i(M^\bullet \otimes^{\mathbf{L}}_R R') \to H^i(M^\bullet \otimes^{\mathbf{L}}_A A') \to H^i(N^\bullet)$$

is an isomorphism. Observe that this is a map of $A \otimes_R R'$-modules, i.e., of A'-modules. On the other hand, (1) is the requirement that for compatible primes $\mathfrak{q}' \subset A'$, $\mathfrak{q} \subset A$, $\mathfrak{p}' \subset R'$, $\mathfrak{p} \subset R$ the composition

$$H^i(M^\bullet_{\mathfrak{q}} \otimes^{\mathbf{L}}_{R_{\mathfrak{p}}} R'_{\mathfrak{p}'}) \otimes_{(A_{\mathfrak{q}} \otimes_{R_{\mathfrak{p}}} R'_{\mathfrak{p}'})} A'_{\mathfrak{q}'} \to H^i(M^\bullet_{\mathfrak{q}} \otimes^{\mathbf{L}}_{A_{\mathfrak{q}}} A'_{\mathfrak{q}'}) \to H^i(N^\bullet_{\mathfrak{q}'})$$

is an isomorphism. Since

$$H^i(M^\bullet_{\mathfrak{q}} \otimes^{\mathbf{L}}_{R_{\mathfrak{p}}} R'_{\mathfrak{p}'}) \otimes_{(A_{\mathfrak{q}} \otimes_{R_{\mathfrak{p}}} R'_{\mathfrak{p}'})} A'_{\mathfrak{q}'} = H^i(M^\bullet \otimes^{\mathbf{L}}_R R') \otimes_{A'} A'_{\mathfrak{q}'}$$

is the localization at $\mathfrak{q}'$, we see that these two conditions are equivalent by Algebra, Lemma 00HN. □